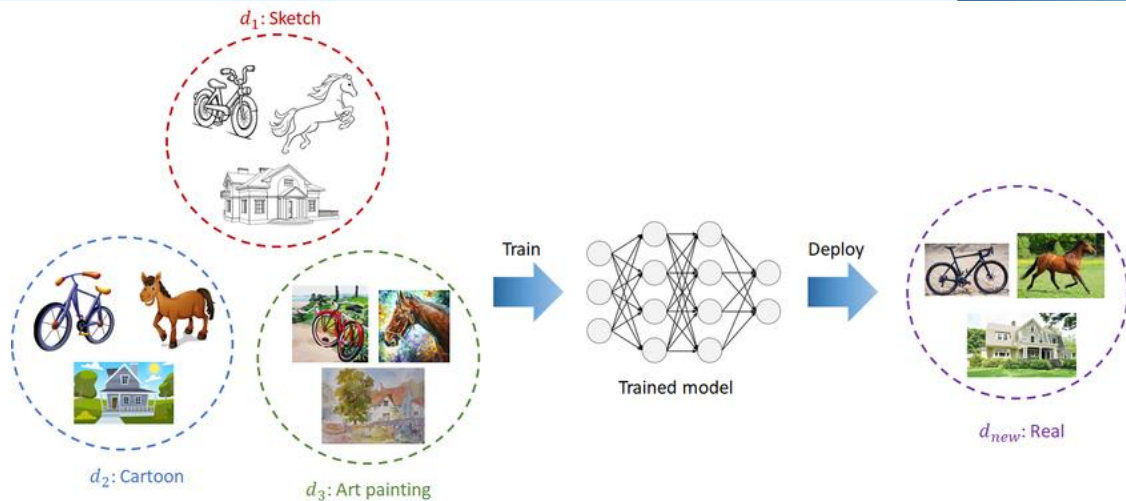


Domain Generalization and introduction to DomainBed



CONTENT

01

The bias problem
in data

02

Introduction to
Domain generalization

03

Domainbed

04

Discussion

Is AI biased?

Searching “CEO” on Google:



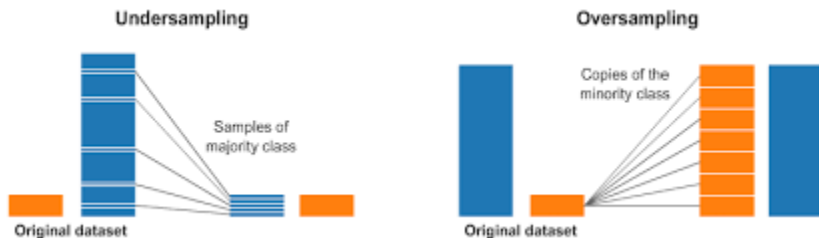
In the past

Similar searches: nurse, programmer, ...



now

Most bias come from your data



Color Matters in Computer Vision

Facial recognition algorithms made by Microsoft, IBM and Face++ were more likely to misidentify the gender of black women than white men.



Gender was misidentified in **up to 1 percent** of **lighter-skinned males** in a set of 385 photos.



Gender was misidentified in **up to 12 percent** of **darker-skinned males** in a set of 318 photos.



Gender was misidentified in **35 percent** of **darker-skinned females** in a set of 271 photos.

When the person in the photo is a **white man**, the software is right **99 percent** of the time.

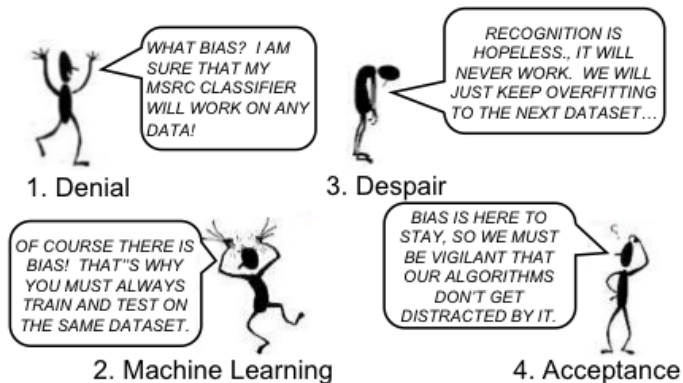
But **the darker the skin**, the **more errors** arise — up to nearly **35 percent** for images of darker skinned women, according to a new study that breaks fresh ground by measuring how the technology works on people of different races and gender.

A more thorough study (CVPR 2011)

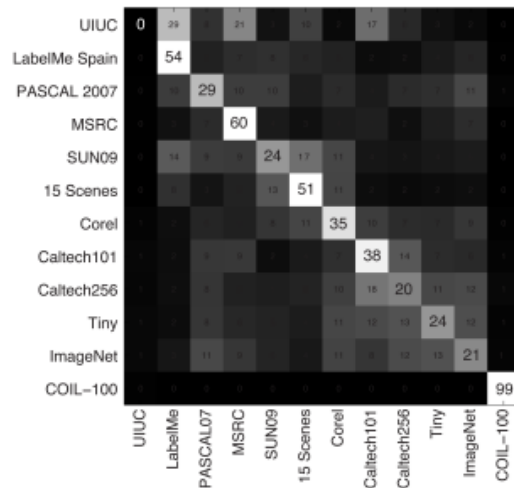
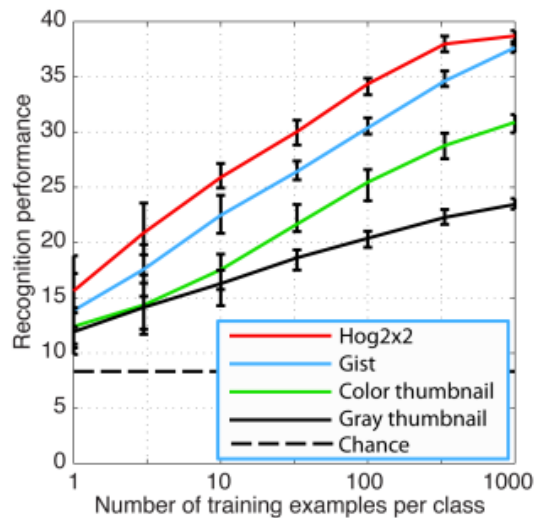
Unbiased Look at Dataset Bias

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Massachusetts Institute of Technology
torralba@csail.mit.edu

Alexei A. Efros
Carnegie Mellon University
efros@cs.cmu.edu



A toy experiment



Chance is $1/12 = 8\%$

PASCAL cars



unusual viewpoints

SUN cars



Caltech101 cars



strong preference for side views

ImageNet cars



racing cars

LabelMe cars



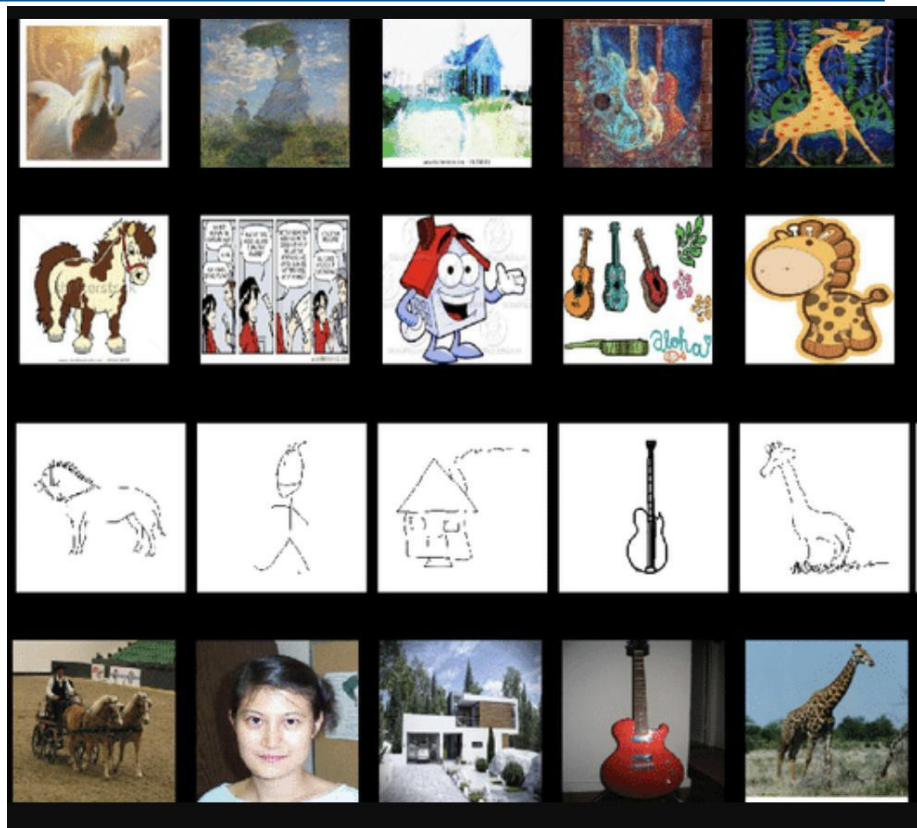
occluded by small object

Key points to consider

- ❑ No report for cross-domain performance in papers
- ❑ Each dataset has its unique, identifiable “signature”
- ❑ Need a solution for the gap between datasets

The problem of data

- ❑ Models do not generalize well to new domains, unlike human
- ❑ Can we collect all the data on the world?
 - ❑ Costly
 - ❑ Regulations
 - ❑ Time



Domain Adaptation

Train on **source** and adapt to **target**

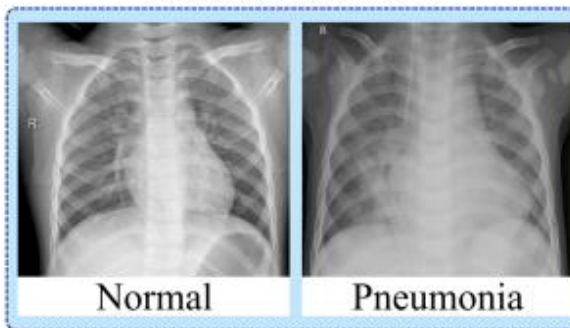


Cityscapes dataset

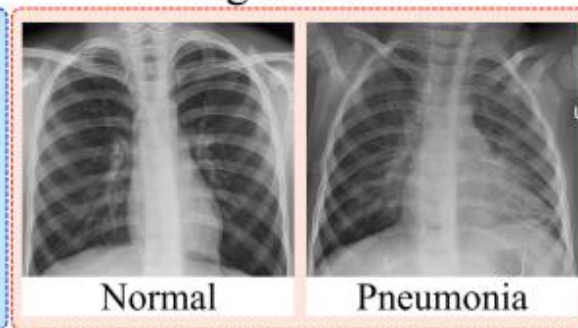


GTA5 dataset

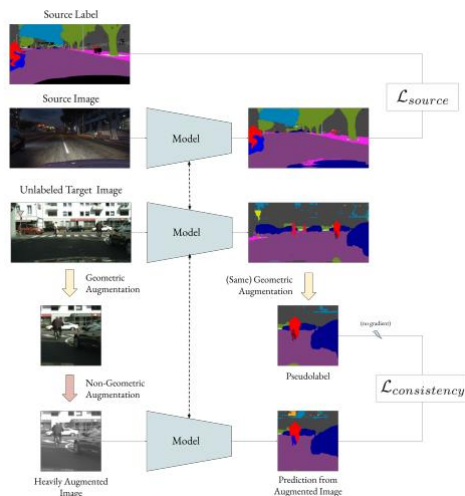
Source domain



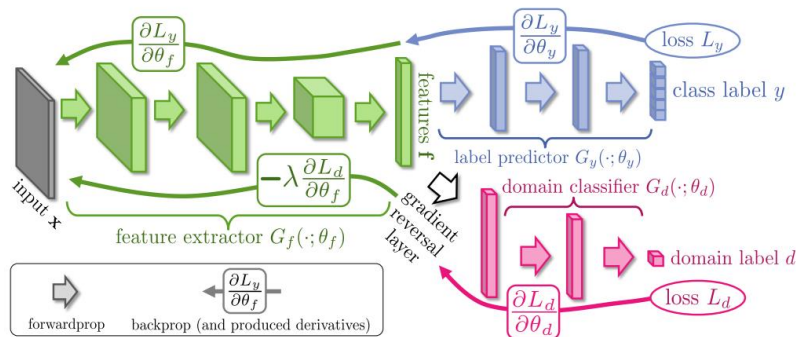
Target domain



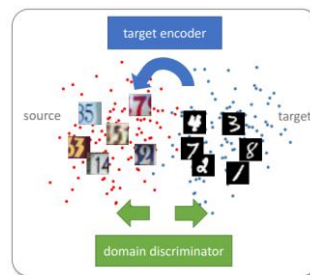
Domain adaptation solutions



Melas-Kyriazi, Luke, and Arjun K. Manrai.
 "Pixmatch: Unsupervised domain adaptation via
 pixelwise consistency training." *Proceedings of
 the IEEE/CVF Conference on Computer Vision
 and Pattern Recognition*. 2021.

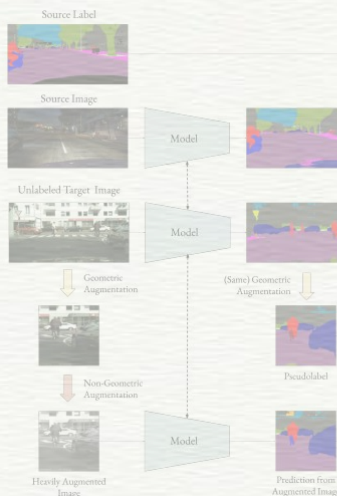


Ganin, Yaroslav, and Victor Lempitsky. "Unsupervised domain adaptation by backpropagation." *International conference on machine learning*. PMLR, 2015.



Tzeng, Eric, et al. "Adversarial discriminative domain adaptation." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017.

Domain adaptation solutions



Melas-Kyriazi, Luke, and Arjun K. Mani. "Pixmatch: Unsupervised domain adaptation by pixelwise consistency training." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2021.

For domain adaptation problems, the target data (unlabeled) is known

Are we able to collect testing data or real data before deployment?

NO!
Privacy Regulations
Real-time deployment

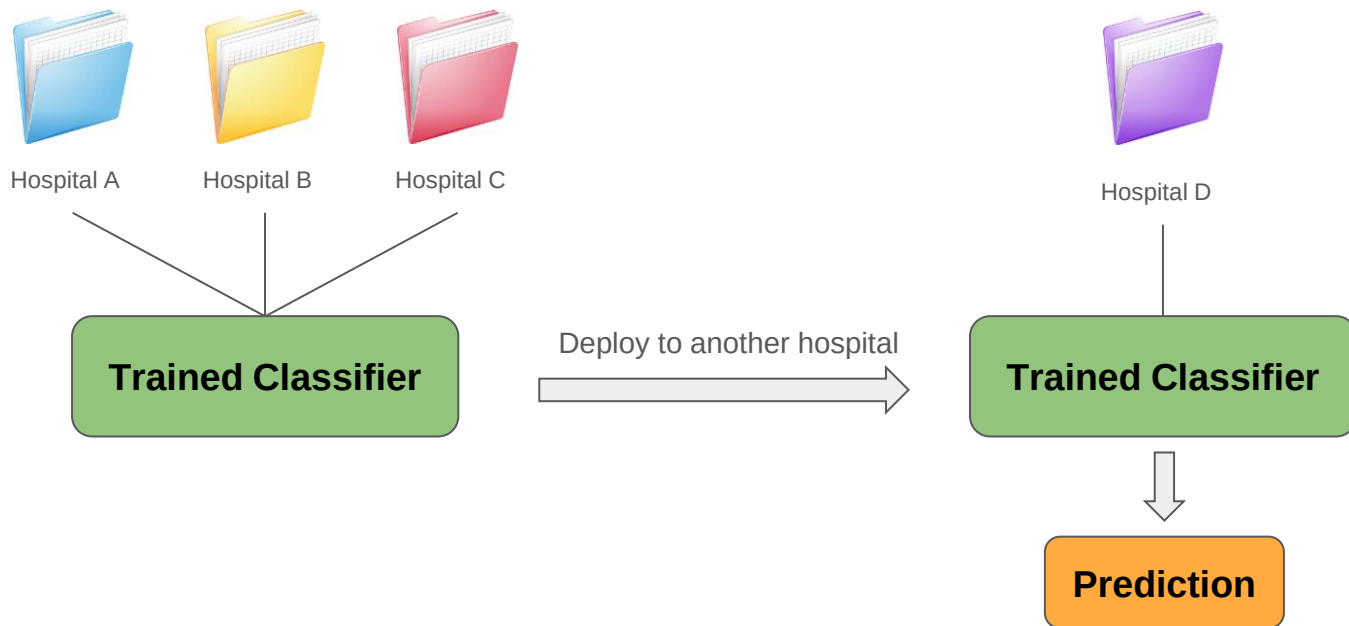
loss L_y
class label y
 $G_d(\cdot; \theta_d)$
domain label d
loss L_d
adaptation by
PMLR, 2015.

domain discriminator

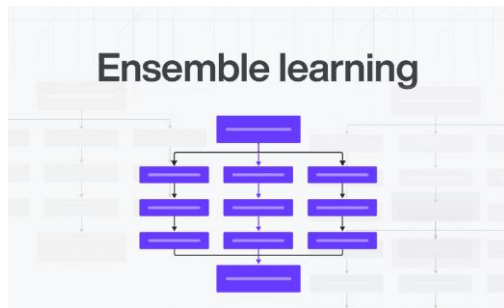
Tzeng, Eric, et al. "Adversarial discriminative domain adaptation." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017.

Domain Generalization

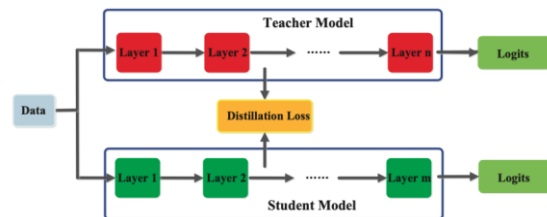
Build a system for previously unseen datasets, given one or multiple training datasets.



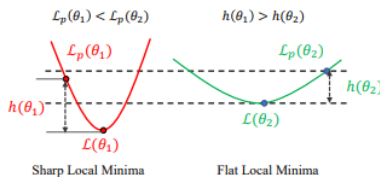
Different approaches



Arpit, Devansh, et al. "Ensemble of averages: Improving model selection and boosting performance in domain generalization." *Advances in Neural Information Processing Systems* 35 (2022): 8265-8277.



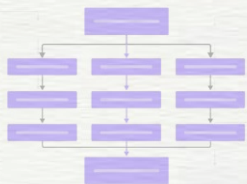
Cha, Junbum, et al. "Domain generalization by mutual-information regularization with pre-trained models." *European conference on computer vision*. Cham: Springer Nature Switzerland, 2022.



Wang, Pengfei, et al. "Sharpness-aware gradient matching for domain generalization." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023.

Different approaches

Ensemble learning



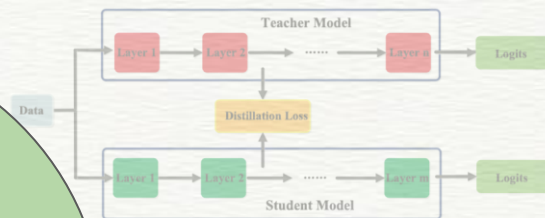
Arpit, Devansh, et al. "Ensemble of averages: Improving model selection and boosting performance in domain generalization." *Advances in Neural Information Processing Systems* 35 (2022): 8265-8277.

A universal platform to evaluate these methods?

- Efficient
- Less complicated
- Accurate
- Reproducible



Wang, Pengfei, et al. "Sharpness-aware gradient matching for domain generalization." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023.



Gunbum, et al. "Domain generalization by mutual-information regularization with pre-trained models." *European conference on computer vision*. Cham: Springer Nature Switzerland, 2022.

Domainbed

Gulrajani, Ishaan, and David Lopez-Paz. "In search of lost domain generalization." *arXiv preprint arXiv:2007.01434* (2020).

1. containing benchmark datasets and algorithms for domain generalization
2. Efficient and less complicated, accurate exp process
3. Reproducible experiments

Dataset	Domains					
Colored MNIST	+90%	+80%	-90%			
	(degree of correlation between color and label)					
Rotated MNIST	0°	15°	30°	45°	60°	75°
VLCS	Caltech101	LabelMe	SUN09	VOC2007		
PACS	Art	Cartoon	Photo	Sketch		
Office-Home	Art	Clipart	Product	Photo		
Terra Incognita	L100	L38	L43	L46		
	(camera trap location)					
DomainNet	Clipart	Infographic	Painting	QuickDraw	Photo	Sketch

Initial release

- 9 algorithms
 - 7 datasets
 - 3 model selection methods
- Empirical Risk Minimization (**ERM**, Vapnik [1998]) minimizes the sum of errors across domains and examples.
 - Group Distributionally Robust Optimization (**DRO**, Sagawa et al. [2019]) performs ERM while increasing the importance of domains with larger errors.
 - Inter-domain Mixup (**Mixup**, Xu et al. [2019], Yan et al. [2020], Wang et al. [2020]) performs ERM on linear interpolations of examples from random pairs of domains and their labels.
 - Meta-Learning for Domain Generalization (**MLDG**, Li et al. [2018a]) leverages MAML [Finn et al., 2017] to meta-learn how to generalize across domains.
 - Different variants of the popular algorithm of Ganin et al. [2016] to learn features $\phi(X^d)$ with distributions matching across domains:
 - Domain-Adversarial Neural Networks (**DANN**, Ganin et al. [2016]) employ an adversarial network to match feature distributions.
 - Class-conditional DANN (**C-DANN**, Li et al. [2018d]) is a variant of DANN matching the conditional distributions $P(\phi(X^d)|Y^d = y)$ across domains, for all labels y .
 - **CORAL** [Sun and Saenko, 2016] matches the mean and covariance of feature distributions.
 - **MMD** [Li et al., 2018b] matches the MMD [Gretton et al., 2012] of feature distributions.
 - Invariant Risk Minimization (**IRM** [Arjovsky et al., 2019]) learns a feature representation $\phi(X^d)$ such that the optimal linear classifier on top of that representation matches across domains.

Dataset	Domains					
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(camera trap location)						
DomainNet	Clipart	Infographic	Painting	QuickDraw	Photo	Sketch
						

<https://github.com/facebookresearch/DomainBed>

